

Multiple Choice

1. **Answer:** A

Options: A) XOR, XNOR, NOR.; B) OR, NOT, AND.; C) NOR, NAND, XNOR.; D) AND, OR.; E) OR, XOR, NOT.

Note: Correct option: A - XOR, XNOR, NOR.

2. **Answer:** A

Options: A) $F(s) = [(-2) / (s + 1)] + [(-8) / (s + 2)]$; B) $F(s) = [(8) / (s + 1)] + [(-10) / (s + 2)]$; C) $F(s) = [(8) / (s + 2)] + [(2) / (s + 1)]$; D) $F(s) = [(-8) / (s + 1)] + [(-2) / (s + 2)]$; E) $F(s) = [(6) / (s + 2)] + [(-14) / (s + 1)]$

Note: Correct option: A - $F(s) = [(-2) / (s + 1)] + [(-8) / (s + 2)]$

3. **Answer:** A

Options: A) $-(2z / ho^2)u_{ho} + (1 / ho^3)u_z$; B) $(z/\rho^3)u_\rho + (1/\rho)u_\theta - (1/\rho^3)u_z$; C) $-(2z / ho^3)u_{ho} + (1 / ho)u_z$; D) $-(z/\rho^2)u_\rho + (1/\rho^3)u_z$; E) $-(1/\rho^2)u_\rho - (2z/\rho^3)u_\theta + (2/\rho)u_z$

Note: Correct option: A - $-(2z / ho^2)u_{ho} + (1 / ho^3)u_z$

4. **Answer:** B

Options: A) 1 %.; B) 2.66 %.; C) 2.5 %.; D) 5 %.; E) 2 %.

Note: Correct option: B - 2.66 %.

5. **Answer:** B

Options: A) $3z^{-2} + 4z^{-1} + 2z^3$; B) $3z^2 + 4z^{-1} + 2z^{-3}$; C) $2z^3 + 4z^{-2} + 3z^{-1}$; D) $4z^2 + 3z^{-1} + 2z^{-3}$; E) $4z^{-3} + 3z^{-2} + 2z^{-1}$

Note: Correct option: B - $3z^2 + 4z^{-1} + 2z^{-3}$

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'V_x = 2 x, V_y = -2 y, V_z = 0'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'i_{forced} = 24 cos 1000 t - 12 sin 1000 t mA'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '20 hp'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $t \sin t, (1 / \sqrt{\pi}) \sin t$ '.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $n! / s^{n+1}$ '.

Long Form Response

11. **Answer:** $f(x) = (1/2) - (2/\pi) \sum_{n=0}^{\infty} (-1)^n [\sin(n)] / (2n + 1)$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** The steam in the container is condensed. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key